

Where is What ?

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List of important Formula

Chapter 1. Co-ordinates of a Point in Space.

p. 1-15

- ✓1. Distance between two given points is

$$[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]^{1/2}. \quad (\text{p. 7})$$
- ✓2. Division of the join of two points

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \quad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, \quad z = \frac{m_1z_2 + m_2z_1}{m_1 + m_2} \quad (\text{p. 11})$$
- ✓3. General co-ordinates of a point on a line joining two given points

$$\left(\frac{x_1 + \lambda x_2}{\lambda + 1}, \frac{y_1 + \lambda y_2}{\lambda + 1}, \frac{z_1 + \lambda z_2}{\lambda + 1} \right) \quad (\text{p. 11})$$

Chapter 2. Direction-Cosines and Projections.

p. 16-36

- ✓1. If l, m, n be the direction-cosines of a line, then

$$l^2 + m^2 + n^2 = 1. \quad (\text{p. 17})$$
- ✓2. If a, b, c be the direction-ratios, then the direction-cosines are

$$\left\{ \frac{a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}} \right\}. \quad (\text{p. 18})$$
- ✓3. Direction-ratios of a line joining two given points are

$$x_2 - x_1, y_2 - y_1, z_2 - z_1. \quad (\text{p. 20})$$
- ✓4. Projection of a line joining two given points on another line whose direction-cosines are given is

$$(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n. \quad (\text{p. 21})$$
- ✓5. Angle between two given lines is given by

$$\cos \theta = l_1l_2 + m_1m_2 + n_1n_2 \quad (\text{p. 22})$$

$$\sin \theta = [(l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2]^{1/2}.$$
6. Lagrange's identity :

$$(l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1n_2)^2$$

$$= (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2. \quad (\text{p. 23})$$
7. Conditions for two lines to be (a) perpendicular (b) Parallel.

$$(a) \quad l_1l_2 + m_1m_2 + n_1n_2 = 0 \quad \text{or} \quad a_1a_2 + b_1b_2 + c_1c_2 = 0 \quad (\text{p. 24})$$

$$(b) \quad l_1 = l_2, m_1 = m_2, n_1 = n_2 \quad \text{or} \quad \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \quad (\text{p. 24})$$

- ✓ 1. Angle between two planes.

$$A_1x + B_1y + C_1z + D_1 = 0, A_2x + B_2y + C_2z + D_2 = 0$$

is $\cos \theta = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{(\sum A_1^2)} \sqrt{(\sum A_2^2)}} \quad (\text{p. 44})$

- ✓ 2. The above planes are parallel or perpendicular according

as $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$

or $A_1A_2 + B_1B_2 + C_1C_2 = 0. \quad (\text{p. 44})$

- ✓ 3. Equation of a plane through the intersection of two planes $P=0, Q=0$ is $P + \lambda Q = 0$ and through a given point is

$$A(x-x_1) + B(y-y_1) + C(z-z_1) = 0. \quad (\text{p. 45})$$

4. Equation of planes in particular cases :

Parallel to co-ordinate planes or perpendicular to co-ordinate axes are $x=\lambda, y=\lambda$ or $z=\lambda. \quad (\text{p. 46})$

5. Perpendicular to co-ordinate planes or parallel to axes are $by + cz + d = 0, ax + cz + d = 0, ax + by + d = 0. \quad (\text{p. 47})$

6. Distance of a point (x', y', z) from

$$Ax + By + Cz + D = 0$$

is $\frac{Ax' + By' + Cz' + D}{\sqrt{A^2 + B^2 + C^2}} \quad (\text{p. 50})$

7. Equations of planes bisecting angles between two given planes are

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}} \quad (\text{p. 62})$$

8. $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ will represent a pair of planes if $abc + 2fgh - af^2 - bg^2 - ch^2 = 0. \quad (\text{p. 66})$

9. If A_x, A_y, A_z be the projections of an area A on the co-ordinate planes, then $A^2 = A_x^2 + A_y^2 + A_z^2. \quad (\text{p. 68})$

- ✓ 1. Equation of a line passing through a given point :

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r \text{ say.}$$

If l, m, n are the actual direction cosines then r is the distance of any point on the line from the given point $(x_1, y_1, z_1).$

(p. 72)

- ✓ Line through two given points is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}. \quad (\text{p. 74})$$

D.R.'s of line

$$\left. \begin{aligned} a_1x + b_1y + c_1z + d_1 &= 0, \\ a_2x + b_2y + c_2z + d_2 &= 0. \end{aligned} \right\}$$
 are given by

$$\left. \begin{aligned} a_1l + b_1m + c_1n &= 0, \\ a_2l + b_2m + c_2n &= 0 \end{aligned} \right\}$$

and any point on it is obtained by putting $z=0$ and solving for x and y . (p. 81)

2. The condition for two given lines to intersect i.e. coplanar is

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0. \quad (\text{p. 104})$$

3. Square of perpendicular distance of a given point (x_1, y_1, z_1)

from the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is

$$\begin{aligned} & \left| \begin{array}{cc} x_1 - \alpha & y_1 - \beta \\ l & m \end{array} \right|^2 + \left| \begin{array}{cc} y_1 - \beta & z_1 - \gamma \\ m & n \end{array} \right|^2 \\ & + \left| \begin{array}{cc} z_1 - \gamma & x_1 - \alpha \\ n & l \end{array} \right|^2 \end{aligned} \quad (\text{p. 121})$$

l, m, n are actual d.c.'s. In case we use D.R.'s a, b, c then divide by $(a^2 + b^2 + c^2)$.

4. Intersection of three planes (1) $\Delta_4 \neq 0$. Planes intersect in a point; (2) $\Delta_4 = 0$, and any of $\Delta_1, \Delta_2, \Delta_3$ is also zero, then they intersect in a line; (3) $\Delta_4 = 0$ and any of $\Delta_1, \Delta_2, \Delta_3$ is not zero, they form a triangular prism. (p. 130)

Chapter 5. Shortest Distance Turn to Page 142-158

Chapter 6. Volume of Tetrahedron p. 159-170

$$V = \frac{1}{6} p \cdot \Delta = \frac{1}{6} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = \frac{1}{6} \begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ x_2 - x_3 & y_2 - y_3 & z_2 - z_3 \\ x_3 - x_4 & y_3 - y_4 & z_3 - z_4 \end{vmatrix}$$

Chapter 7. Skew Lines p. 171-188

Equations to two skew lines by proper choice of axes are
 $y = x \tan \alpha, z = c; y = -x \tan \alpha, z = -c$. (p. 172)

or $\frac{x}{1} = \frac{y}{m} = \frac{z-c}{0}; \frac{x}{1} = \frac{y}{-m} = \frac{z+c}{0}$ where $m = \tan \alpha$.

1. In order to shift the origin to (α, β, γ) without changing the direction of axes, replace x, y and z by $x+\alpha, y+\beta, z+\gamma$ respectively. (p. 189)
2. If (x, y, z) are the co-ordinates of point P referred to co-ordinate axes and (x', y', z') , be the corresponding co-ordinates referred to another set of rectangular axes through the same origin and having direction-cosines $l_r, m_r, n_r, r=1, 2, 3$, then

$$x = l_1 x' + l_2 y' + l_3 z', \quad y = m_1 x' + m_2 y' + m_3 z'$$

and

$$z = n_1 x' + n_2 y' + n_3 z'$$

or

$$x' = l_1 x + m_1 y + n_1 z, \quad y' = l_2 x + m_2 y + n_2 z,$$

$$z' = l_3 x + m_3 y + n_3 z.$$

(p. 190)

Easy way to remember :—

	x'	y'	z'	New
x	l_1	l_2	l_3	d.c.'s
y	m_1	m_2	m_3	d.c.'s
z	n_1	n_2	n_3	d.c.'s
Old d.c.'s d.c.'s d.c.'s				

(p. 190)

In the above chart horizontal lines as well as vertical lines denote the d.c.'s of mutually perpendicular lines and hence we have relations A, B, C, D to follow in (3).

To get the value of x multiply each element of this row with the corresponding element of first row and add i.e.

$$x = l_1 x' + l_2 y' + l_3 z'.$$

Similarly to find the value of z' , multiply each element of this column with the corresponding element of 1st column and add i.e. $z' = l_3 x + m_3 y + n_3 z$ etc.

3. Relation between the direction-cosines of three mutually perpendicular lines;

$$l_1^2 + m_1^2 + n_1^2 = 1, \quad \Sigma l_2^2 = 1, \quad \Sigma l_3^2 = 1$$

...(A)

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0, \Sigma l_2 l_3 = 0, \Sigma l_3 l_1 = 0. \quad \dots(B)$$

$$l_1^2 + l_2^2 + l_3^2 = 1, \Sigma m_1^2 = 1, \Sigma n_1^2 = 1. \quad \dots(C)$$

$$l_1 m_1 + l_2 m_2 + l_3 m_3 = 0, \Sigma m_1 n_1 = 0, \Sigma n_1 l_1 = 0. \quad \dots(D) \quad (p. 191)$$

Chapter 9.**Sphere**

(p. 196–245)

1. The centre and radius of the sphere
 $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$
 are $(-u, -v, -w)$ and $\sqrt{(u^2 + v^2 + w^2 - d)}$ (p. 197)

2. Sphere on the join of two given points as diameter,
 $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) + (z-z_1)(z-z_2) = 0$. (p. 199)

3. A sphere and a plane or two spheres taken together represent a circle and the equation to the sphere through this circle are $S + \lambda P$ or $S + \lambda S' = 0$.

The equation of sphere $OABC$ is

$$x^2 + y^2 + z^2 - ax - by - cz = 0. \quad (p. 200)$$

where A, B, C are $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$.

4. The equation of the tangent plane at (x_1, y_1, z_1) is

$$xx_1 + yy_1 + zz_1 + u(x+x_1) + v(y+y_1) + w(z+z_1) + d = 0$$

and it is perpendicular to the radius through the point of contact. (p. 221)

5. Condition for orthogonal intersection of two spheres is

$$2u_1 u_2 + 2v_1 v_2 + 2w_1 w_2 = d_1 + d_2. \quad (p. 234)$$

6. The square of the length of the tangent from a given point to a given sphere in whose equation, the coefficient of x^2, y^2 and z^2 are each unity is obtained by substituting the co-ordinates of the point in its equation. (p. 239)

7. The radical plane of two given spheres is

$$2x(u_1 - u_2) + 2y(v_1 - v_2) + 2z(w_1 - w_2) + d_1 - d_2 = 0.$$

$$\text{i.e.,} \quad S_1 - S_2 = 0. \quad (p. 240)$$

8. Coaxial sphere $x^2 + y^2 + z^2 + 2ux + d = 0$,

where d is constant and u variable. (p. 242)

Chapter 10.**Cone**

(p. 246–278)

1. If $x/l = y/m = z/n$ is a generator of a cone $f(x, y, z) = 0$ (homogeneous) then $f(l, m, n) = 0$. (p. 247)

2. The general equation of cone through the axis is

$$fyz + gzx + hxy = 0. \quad (p. 248)$$

3. The equation of tangent plane of the cone $f(x, y, z) = 0$ at (α, β, γ) is

$$x \frac{\partial f}{\partial \alpha} + y \frac{\partial f}{\partial \beta} + z \frac{\partial f}{\partial \gamma} = 0 \quad (\text{p. 263})$$

4. The equation of a cone reciprocal to $ax^2+by^2+cz^2=0$ is $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} = 0$ i.e. replace the coefficients by their reciprocals.

(p. 267)

5. The condition for the plane $ux+vy+wz=0$ to cut the cone $f(x, y, z)=0$ in perpendicular lines is

$$(a+b+c)(u^2+v^2+w^2)=f(u, v, w). \quad (\text{p. 271})$$

6. Condition for the cone $f(x, y, z)=0$ to have three mutually perpendicular generators is $a+b+c=0$ and will have three mutually perpendicular tangent planes if $A+B+C=0$ or $ab+bc+ca=f^2+g^2+h^2$.

(p. 281)

Chapter 11. Right circular cone and cylinder (p. 288—300)

Chapter 12. The Conicoids (p. 301—325)

1. Condition for the plane $lx+my+nz=p$ to touch the conicoid $ax^2+by^2+cz^2=1$ is $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$ and the point of contact is $\left(\frac{l}{ap}, \frac{m}{bp}, \frac{n}{cp}\right)$. For the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, the condition is

$$a^2l^2+b^2m^2+c^2n^2=p^2. \quad (\text{p. 307})$$

2. The polar plane is $a\alpha x+b\beta y+c\gamma z=1$.

Polar line of a given line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ is

$$\begin{aligned} a\alpha x+b\beta y+c\gamma z-1 &= 0, \\ alx+bmy+cnz &= 0. \end{aligned} \quad (\text{p. 315})$$

3. The locus of the chords of the conicoid bisected at (α, β, γ) is $a\alpha x+b\beta y+c\gamma z=a\alpha^2+b\beta^2+c\gamma^2$ or $T=S_1$ and the section of the conicoid by this plane is a conic whose centre is (α, β, γ) .

(p. 320)

Chapter 13. Enveloping Cone and Enveloping Cylinder (p. 326—336)

1. Enveloping cone is the locus of tangent lines from a given point to a given surface and its equation is

$$SS_1=T^2. \quad (\text{p. 326})$$

2. Enveloping cylinder is the locus of the tangents to a surface which are parallel to a given line.

(p. 331)

1

Co-ordinates of a Point in Space

Introduction. Students are very well acquainted with the position of a point in a plane which is referred to two intersecting straight lines called the **axes of reference** and their point of intersection called the **origin** of co-ordinates. If the angle between the axes be a right angle, then they are termed as **rectangular axes**, otherwise **oblique**, and these axes divide the plane into **four quadrants** called first, second, third and fourth quadrants respectively.

In Fig. 1, if P_1 be any point and we draw P_1M_1 perpendicular on $X'OX$ called x -axis, then $OM_1 = x$ measures the x -co-ordinate called **abscissa** and $M_1P_1 = y$ measures the y -co-ordinate called **ordinate** of that point.

These two numbers together are called the **co-ordinates** of the point and are written as (x, y) . You are also acquainted with the convention of sign *i.e.*,

“Plus to the right, minus to the left ;
Positive height, negative depth.”

Thus with the above convention the x -co-ordinate *i.e.*, abscissa OM_2 of the point P_2 is negative whereas its ordinate P_2M_2 is +ve. Similarly the point P_3 in the third quadrant has both its abscissa and ordinate negative. In case the axes be oblique, then from P_1 , we draw P_1M_1 parallel to y -axis meeting the x -axis in M_1 ; then OM_1 is the abscissa and M_1P_1 is the ordinate. (See Fig. 2 next page)

Now the question arises if it is possible for us to represent

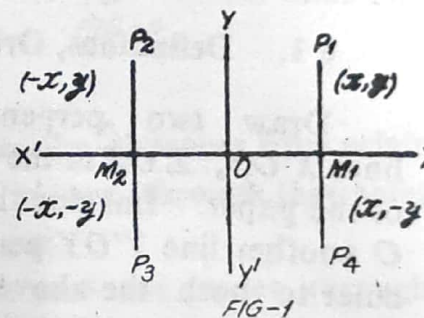


FIG-1

all the points we can imagine with reference to the above system. Yes, we can, provided all the points lie in the same plane which is not possible. For example any five corners of your study room do not lie in the same plane and hence we cannot represent

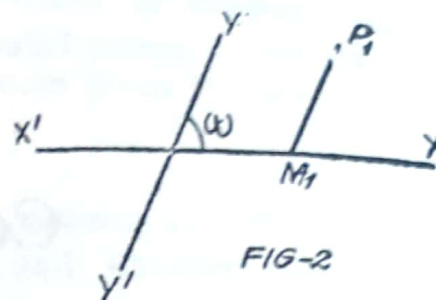


FIG-2

them by the above system. Similarly all those bodies which occupy space e.g. a box, cannot be described completely by a plane, for it does not wholly lie in it. It has three dimensions i.e. length, breadth and depth. The geometry of all such bodies which occupy space, i.e. of three dimensions, is called the geometry of three dimensions and is also known as solid geometry. One thing should be clearly understood here that by solid we do not mean the absence of 'hollowness'. When we speak of a cone in solid geometry we only mean a hollow cone of negligible thickness. Similarly by a sphere we mean hollow sphere. The outer boundaries of these solids are called surface. Hence in the solid geometry we shall discuss about the position of a point in space.

§ 1. Definitions, Origin, axes and co-ordinate planes.

Draw two perpendicular lines $X'OX$, $Z'OZ$ in the plane of the paper. Imagine through O another line $Y'OY$ perpendicular to both the above lines, OY pointing towards the reader. The set of these three mutually perpendicular lines is called the axes of co-ordinates (rectangular). In case they be not

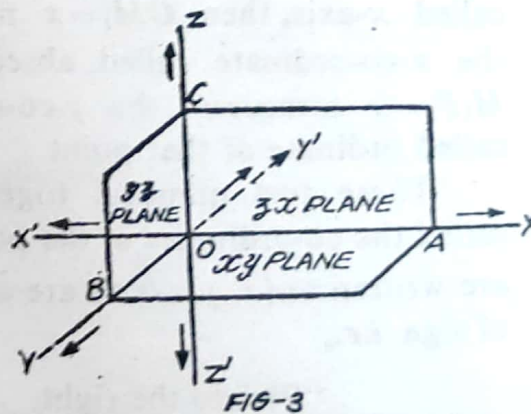


FIG-3

mutually perpendicular then they are termed as oblique axes. We shall however deal in this book with rectangular axes and it will be understood, that the axes are rectangular unless otherwise stated. The point O is called the origin. The positive and negative directions of these axes are noted in the figure. These three axes taken in pairs give us three planes YOZ , ZOX and XOY also called as yz , zx and xy planes respectively. These three planes are called rectangular co-ordinate planes.

Hence we conclude that co-ordinates of a point P are respectively the distances of the point P from the three co-ordinate planes.

(c) Again plane $PMAN$ is parallel to the yz -plane and therefore perpendicular to x -axis. Since PA is any line in this plane, it is therefore perpendicular to OX . Similarly PB is perpendicular to OY and PC is perpendicular to OZ .

Hence

x -co-ordinate $= OA$, where A is the foot of the perpendicular from P on x -axis.

y -co-ordinate $= OB$, where B is the foot of the perpendicular from P on y -axis.

z -co-ordinate $= OC$, where C is the foot of the perpendicular from P on z -axis.

Hence we conclude that the co-ordinates of a point P are respectively the distances from O of the feet of the perpendiculars A , B , C from P on the axes respectively.

(b) Again $NP = AM = OC = z$ -co-ordinate $= c$.

$AN = OB = y$ -co-ordinate $= b$.

$ON = x$ -co-ordinate $= a$.

Hence if from any point P we draw PN perpendicular to xy -plane and from N the foot of the perpendicular NA be drawn parallel to the y -axis meeting x -axis in A , then

$OA = a$, $AN = b$, $NP = c$. (see fig. 4 on page 3)

Converse. If we are given the three numbers a, b, c , then to find the point whose co-ordinates are these numbers we shall follow the method given below.

Measure along the axes respectively OA, OB, OC equal to the given numbers a, b, c respectively. Now through the point A, B, C , draw planes, parallel respectively to the three co-ordinate planes i.e. yz, zx and xy -planes. Then the point P where these three planes meet is the required point whose co-ordinates are the given numbers a, b, c .

Octants. The three co-ordinate planes divide the whole of the space into eight parts which are called octants. The octant $OXYZ$ in which the point P is situated is called first octant and any point in this has each of its co-ordinates positive. Again since each of co-ordinates of the point P may be either positive or negative, we shall have eight such points $P(a, b, c)$ whose co-ordinates may be represented numerically by three numbers a, b

and c but have different signs. Below we give eight octants and the co-ordinates of the points P , in each of them situated with respect to the axes just as P is situated in the octant $OXYZ$ and has co-ordinates (a, b, c) .

- | | |
|---------------------------|-----------------------------|
| 1. $OXYZ (a, b, c)$. | 2. $OX'YZ (-a, b, c)$. |
| 3. $OXY'Z (a, -b, c)$. | 4. $OXYZ' (a, b, -c)$. |
| 5. $OXY'Z' (a, -b, -c)$. | 6. $OX'YZ' (-a, b, -c)$. |
| 7. $OX'Y'Z (-a, -b, c)$. | 8. $OX'Y'Z' (-a, -b, -c)$. |

Note. If the octant contains the negative side of either axis, then the co-ordinate corresponding to that is also —ive e.g. a point in the octant $OX'Y'Z'$ will have both y and z -co-ordinates negative.

Example 1. If the co-ordinates of the point P in fig. 4 are (a, b, c) , write down co-ordinates of the points A, B, C, L, M, N . Also write down the general co-ordinates of a point in three co-ordinate planes.

Since the co-ordinates of the point P are (a, b, c) we have
 $OA=a, OB=b, OC=c$.

A being a point on the x -axis has co-ordinates $(a, 0, 0)$.

Similarly the point B is $(0, b, 0)$ and C is $(0, 0, c)$.

Again L being a point in the yz -plane has x -co-ordinate zero because x -co-ordinate of a point is its distance from the yz -plane measured from the point parallel to x -axis. Its y -co-ordinate is its distance from zx -plane and it is therefore $CL=OB=b$. Its z -co-ordinate is the distance from the xy -plane $=LB=OC=c$. Hence the point L is $(0, b, c)$.

Similarly the point M being in zx -plane is $(a, 0, c)$ and N being in xy -plane is therefore the point $(a, b, 0)$. Now we can say that the co-ordinates of a point in the xy -plane are $(x, y, 0)$, in yz -plane $(0, y, z)$ and zx -plane $(x, 0, z)$.

Example 2. What are the situations of the following points.

- (a) $(-1, -2, -3)$, (b) $(0, -2, 0)$, (c) $(-1, -2, 0)$,
 (d) $(0, 0, 3)$, (e) $(1, -2, -3)$.

(a) It is a point in the octant $OX'Y'Z'$, such that its distance from the co-ordinate planes $y'z'$, zx and xy are respectively, 1, 2 and 3.

(b) It is a point on the —ive side of y -axis situated at a distance 2 from the origin.

(c) It is a point in the octant $OX'Y'Z$ and is in the co-ordinate plane $X'Y'$ as z -co-ordinate is zero.

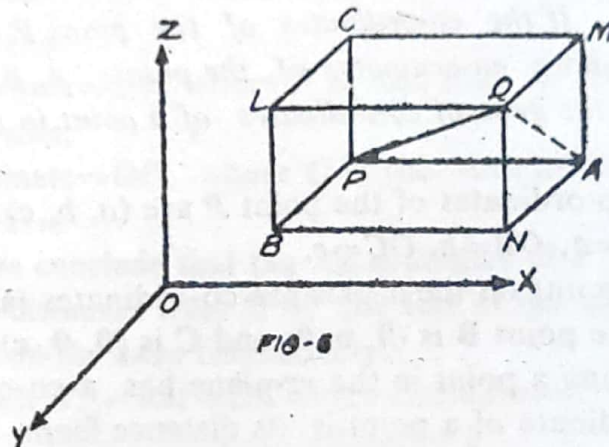
(d) It is a point on the positive side of z -axis situated at a distance 3 from the origin.

(e) It is a point in the octant $OXY'Z'$, such that its distance from the three co-ordinate planes are 1, 2 and 3 respectively.

§ 3. Distance between two points.

1st Method.

Let the co-ordinates of the two points P and Q be (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively. Through P and Q draw planes $PBLC$ and $ANQM$ parallel to yz -plane, planes $APCM$ and $NBLQ$ parallel to zx -plane and planes $ANBP$ and $MQLC$ parallel to xy -plane. Now from the definition of co-ordinates of a point, x -co-ordinate of a point is its distance from the yz -plane. Hence the distance



between the two planes $PBLC$ and $ANQM$ each parallel to the yz -plane is the difference of their x -co-ordinates,

$$\therefore PA = x_2 - x_1.$$

Similarly $AN = y_2 - y_1$ and $NQ = z_2 - z_1$ (1)

Now QN being perpendicular to xy -plane is also perpendicular to plane $ANBP$ which is parallel to xy -plane; therefore $\angle QNA = \pi/2$.

Hence from the right-angled triangle QNA , we have

$$AQ^2 = AN^2 + NQ^2. \quad \dots (2)$$

Again QA is any line in the plane $ANQM$ which is perpendicular to the line PA ; hence QA is perpendicular to PA . Therefore from right angled triangle QAP , we get

$$PQ^2 = PA^2 + AQ^2. \quad \dots (3)$$

Substituting in (3) the value of AQ^2 from (2), we get

$$PQ^2 = PA^2 + AN^2 + NQ^2$$

or $PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$ from (1).

Distance between Two Points

7

Second Method.

Through P and Q draw PM and QN perpendiculars to the xy -plane, so that the co-ordinates of the feet of these perpendiculars M and N are $(x_1, y_1, 0)$ and $(x_2, y_2, 0)$ respectively. The plane co-ordinates of these two points in the xy -plane referred to OX and OY as axes are (x_1, y_1) and (x_2, y_2) and hence by Plane Co-ordinate Geometry we have

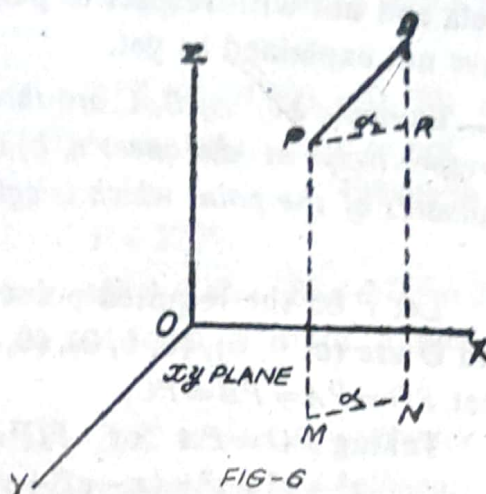


FIG-6

$$MN^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2. \quad \dots(1)$$

Now PM and QN are clearly parallel and hence lie in the same plane. Therefore if we draw a line through P parallel to MN , then it also lies in this plane and hence meets QN in the point R and is also perpendicular to it.

Now from right-angled $\triangle PQR$, we get

$$\begin{aligned} PQ^2 &= PR^2 + QR^2 \\ &= MN^2 + (QN - RN)^2 \quad (\because PR = MN) \\ &= MN^2 + (QN - PM)^2 \end{aligned}$$

Now $PM = z_1$ and $QN = z_2$;

$$\therefore PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 \text{ from (1) ;}$$

$$\therefore PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]} \text{ in magnitude.}$$

Cor. In case we are to find the distance of given point from the origin then putting $x_2 = y_2 = z_2 = 0$, we get the distance of the point (x_1, y_1, z_1) from origin as

$$\sqrt{(x_1^2 + y_1^2 + z_1^2)}.$$

Note : Properties of Geometrical figures,

Isosceles triangle Two sides equal

Equilateral triangle All sides equal

right angled triangle Sum of the squares on two sides is equal to the square on the third

Square All sides equal, Diagonal equal

Rhombus All sides equal, Diagonals not equal

Rectangle Opposite sides equal, Diagonals equal

Parallelogram Opposite sides equal, Diagonals not equal.

Sphere Distances of every point from the centre are equal.

We have given the properties only in relation to distance formula and not with respect to perpendicularity condition which we have not explained as yet.

Example 1. A, B, C are three points on the axis of x, y and z respectively at distances a, b, c from the origin O ; find the co-ordinates of the point which is equidistant from A, B, C and O .

(Gaubati 69)

Let P be the required point (x, y, z) and the point A, B, C and O are $(a, 0, 0), (0, b, 0), (0, 0, c)$ and $(0, 0, 0)$; we are given that $PO = PA = PB = PC$.

Taking $PO = PA$ or $PO^2 = PA^2$, we get

$$x^2 + y^2 + z^2 = (x - a)^2 + y^2 + z^2$$

$$0 = a^2 - 2ax \text{ i.e. } x = a/2$$

Similarly taking $PO^2 = PB^2$ and $PO^2 = PC^2$, we get

$$y = b/2 \text{ and } z = c/2.$$

Hence the required point P is $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$.

Example 2. (a) Find the locus of a point which moves so that the sum of its distances from the points $(a, 0, 0)$, and $(-a, 0, 0)$ is constant.

(Gaubati 69)

Let the point be P and we are given that $PA + PB = 2k$ say

$$\therefore \sqrt{(x-a)^2 + y^2 + z^2} + \sqrt{(x+a)^2 + y^2 + z^2} = 2k$$

$$\text{or } L + M = 2k \quad \dots (1)$$

$$\text{But } L^2 - M^2 = (x-a)^2 - (x+a)^2 = -4ax \quad \text{Divide}$$

$$L - M = \frac{-2ax}{k} \quad \dots (2) \quad \text{Adding (1) and (2)}$$

$$2L = \left(k - \frac{ax}{k}\right) \quad \text{Square} \quad L^2 = \left(k - \frac{ax}{k}\right)^2$$

$$\text{or } (x-a)^2 + y^2 + z^2 = k^2 - 2ax + \frac{a^2x^2}{k^2}$$

$$\text{or } x^2 \left(1 - \frac{a^2}{k^2}\right) + y^2 + z^2 = k^2 - a^2 \text{ is the required locus.}$$

(b) Prove that the locus of the point which moves so that the differences of its distances from the points $(2, 0, 0)$, $(-2, 0, 0)$ is 60 $x^2 - 4(y^2 + z^2) = 15$.

(Punjab 75)

Exercise 1

1. Prove that the triangle formed by joining the three points whose co-ordinates are $(1, 2, 3)$, $(2, 3, 1)$ and $(3, 1, 2)$ respectively is an equilateral triangle.

Here clearly $AB=BC=CA=\sqrt{6}$. Hence Δ is equilateral.

✓2. (a) Show that the points $(0, 7, 10)$, $(\frac{1}{4}, 6, 6)$, $(-4, 9, 6)$ form an isosceles right angled triangle. (Raj. 73, Utkal 77)

Here $AB=BC=\sqrt{18}$ and $CA=\sqrt{36}$. Also $AB^2+BC^2=18+18=36=CA^2$.

(b) Verify whether the points $A(3, 6, 9)$, $B(10, 20, 30)$ and $C(25, -41, 5)$ are the vertices of a right angled triangle or not. (Magadh 72)

Here $AB^2=686$, $BC^2=4571$, $CA^2=2709$.

Now BC^2 is the greatest but $AB^2+CA^2=686+2709=3395 \neq BC^2$. Hence the given points do not form a right angled triangle.

✓3. Prove that the three points A, B, C whose co-ordinates are $(3, -2, 4)$, $(1, 1, 1)$ and $(-1, 4, -2)$ respectively are collinear.

Here $AB=\sqrt{22}$, $BC=\sqrt{22}$, $AC=\sqrt{88}=2\sqrt{22}$.

$\therefore AB+BC=AC$ and hence collinear.

4. are written the co-ordinates of four points A, B, C and D . Prove that they enclose the figure written against each.

(i) $A(5, -1, 1)$ $B(7, -4, 7)$

$C(1, -6, 10)$ $D(-1, -3, 4)$

Rhombus

(ii) $A(1, 1, 1)$, $B(-2, 4, 1)$

$C(-1, 5, 5)$ $D(2, 2, 5)$

Square

(iii) $A(1, 2, 3)$, $B(-1, -2, -1)$

$C(2, 3, 2)$ $D(4, 7, 6)$

parallelogram

Use the properties given on Page 7.

✓5. Find the centre of the sphere through the four points $(0, 0, 0)$, $(2, 0, 0)$, $(0, 4, 0)$, $(0, 0, 6)$ and also find its equation.

Let the centre be (x, y, z) and radius be r then the distance of centre from each of the four given points is equal to radius r .

$$\therefore x^2+y^2+z^2=r^2 \quad \dots(1)$$

$$(x-2)^2+y^2+z^2=r^2 \quad \dots(2)$$

$$x^2+(y-4)^2+z^2=r^2 \quad \dots(3)$$

$$x^2+y^2+(z-6)^2=r^2 \quad \dots(4)$$

Subtracting 1st from each of (2), (3) and (4), we get

$$(x-2)^2-x^2=0 \quad \text{or} \quad -4x+4=0 \quad \therefore x=1$$

$$(y-4)^2-y^2=0 \quad \therefore -8y+16=0 \quad \therefore y=2$$

$$(z-6)^2-z^2=0 \quad \therefore -12z+36=0 \quad \therefore z=3$$

Hence centre is $(1, 2, 3)$ and from (1), we get

$$1+4+9=r^2 \quad \therefore r=\sqrt{14}.$$

Hence the equation of the sphere is

$$(x-1)^2 + (y-2)^2 + (z+3)^2 = (\sqrt{14})^2$$

or

$$x^2 + y^2 + z^2 - 2x - 4y - 6z = 0.$$

6. A and B are two points (0, 2, 3) and (2, -2, 1) respectively. Find the locus of a point P such that

- (i) it is equidistant from the two given points :
- (ii) the sum of the squares of its distances from the given points is constant; equal to k^2 .

Ans. (i) $x - 2y - z + 1 = 0$

(ii) $x^2 + y^2 + z^2 - 2x - 4z + 11 - k^2 = 0$

Example 7. (a) Find the locus of a point P which moves so that its distances from the point A (-2, 2, 3) and B (13, -3, 13) are governed by the relation $3PA = 2PB$. $9PA^2 = 4PB^2$ etc.

$$x^2 + y^2 + z^2 + 28x - 12y + 10z - 247 = 0.$$

(b) Prove that the locus of a point the difference of whose distances from the points A (0, 0, 4) and B (0, 0, -4) is 4 is

$$x^2 + y^2 - 3z^2 + 12 = 0$$

[Punjab 73]

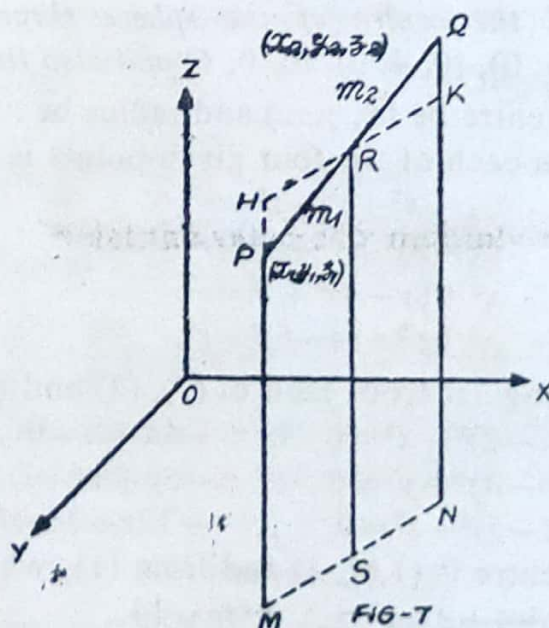
(c) Prove that the locus of a point the sum of whose distances from (4, 0, 0) and (-4, 0, 0) is 10 is $9x^2 + 25(y^2 + z^2) = 225$.

[Bihar 76]

Proceed as in solved Ex. 2 page 8.

§ 4. Division of the join of two points. To find the co-ordinates of a point which divides the line joining two given points P (x_1, y_1, z_1) and Q (x_2, y_2, z_2) in a given ratio $m_1 : m_2$.

Let R be the point (x, y, z) which divides the join of P and Q



in the ratio $m_1 : m_2$. From P, R and Q draw perpendiculars PM,

RS and QN to the xy -plane. These three lines being all perpendicular to the xy -plane are parallel and since they are cut by the same line PQR , they lie in the same plane. Through R draw a line parallel to MSN and this line also lies in the same plane and hence it cuts MP produced in H and NQ in K .

$$\therefore \frac{m_1}{m_2} = \frac{PR}{RQ} = \frac{PH}{KQ} = \frac{HM-PM}{QN-KS} = \frac{RS-PM}{QN-RS} = \frac{z-z_1}{z_2-z}$$

$$\text{or } m_1 z_2 - m_1 z = m_2 z - m_2 z_1$$

$$\text{or } z(m_1 + m_2) = m_1 z_2 + m_2 z_1$$

$$\text{or } z = \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2}$$

Similarly by drawing perpendiculars on zx and yz -planes and proceeding as above, we can prove that

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \text{ and } x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

Note. If $\frac{m_1}{m_2}$ is +ive, then the point R divides the line PQ internally and if it is -ive, then externally.

Cor. 1. Co-ordinates of the mid-point.

If R be the middle point of the line PQ , then $m_1 : m_2 : 1 : 1$ and hence from the last formulae the co-ordinates of R are

$$x = \frac{1}{2}(x_1 + x_2), y = \frac{1}{2}(y_1 + y_2) \text{ and } z = \frac{1}{2}(z_1 + z_2).$$

Cor. 2. General co-ordinates of a point on the line PQ .

If we choose $m_1 : m_2 :: \lambda : 1$ i.e. $\frac{m_1}{m_2} = \lambda$, then the co-ordinates of the points R are

$$\left[\frac{x_1 + \lambda x_2}{\lambda + 1}, \frac{y_1 + \lambda y_2}{\lambda + 1}, \frac{z_1 + \lambda z_2}{\lambda + 1} \right].$$

For every point on the line PQ there corresponds a value of λ and for every value of λ we shall get a point on the line PQ .

§ 5. Centre of gravity of a triangle and a tetrahedron.

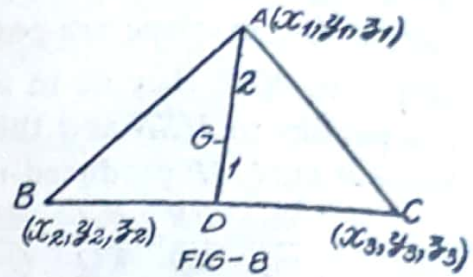
To prove that the c.g. of a triangle, the co-ordinates of whose vertices are the points (x_r, y_r, z_r) $r=1, 2, 3$, is the point

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

The co-ordinates of D , the mid-point of BC , are

$$\left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}, \frac{z_2 + z_3}{2} \right)$$

Now we know from geometry that the C.G. of a triangle is a point on the median dividing it in the ratio 2 : 1. Hence if $G(x, y, z)$ be the C.G. of the $\triangle ABC$ then



$$x = \frac{2 \cdot \frac{x_2 + x_3}{2} + 1 \cdot x_1}{2 + 1} = \frac{x_1 + x_2 + x_3}{3}$$

$= \frac{1}{3}$ (sum of the x -co-ordinates of the vertices).

$$\text{Similarly } y = \frac{y_1 + y_2 + y_3}{3}, \quad z = \frac{z_1 + z_2 + z_3}{3}.$$

Tetrahedron :

To prove that the centre of gravity of a tetrahedron, the co-ordinates of whose vertices are (x_r, y_r, z_r) , $r = 1, 2, 3, 4$, is the point.

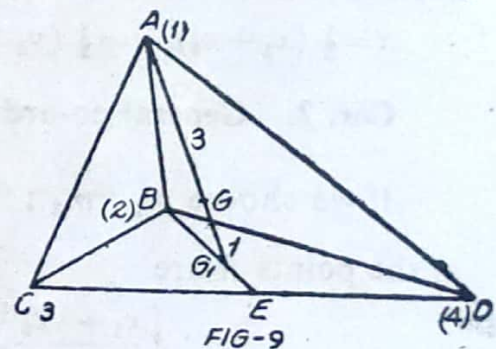
$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

[Rohailkhand 79]

Let BCD be a triangle and A another point not in the plane determined by BCD . Join AB, AC, AD ; then the four faces of the tetrahedron $ABCD$ are the triangles BCD, DBA, ABC and ACD and the four vertices are the points A, B, C, D . It has six edges which in pairs of opposite edges are

$$(AB, CD), (BC, AD), (DB, AC).$$

If G_1 be the C.G. of the triangular face BCD , then the point G_1 is



$$\left(\frac{x_2 + x_3 + x_4}{3}, \frac{y_2 + y_3 + y_4}{3}, \frac{z_2 + z_3 + z_4}{3} \right)$$

Now we know from geometry that the C. G. of a tetrahedron is on the line joining any vertex to the C. G. of the opposite face and divides it in the ratio 3 : 1.

Hence if $G(x, y, z)$ be the C. G. of the tetrahedron then

$$x = \frac{3 \frac{x_2 + x_3 + x_4}{3} + 1 \cdot x_1}{3 + 1} = \frac{x_1 + x_2 + x_3 + x_4}{4}$$

$= \frac{1}{4}$ (sum of the x -co-ordinates of the four vertices).

$$\text{Similarly } y = \frac{y_1 + y_2 + y_3 + y_4}{4}, \quad z = \frac{z_1 + z_2 + z_3 + z_4}{4}.$$

Example 2 Find the ratio in which the yz -plane divides the join of the points $(-2, 4, 7)$ and $(3, -5, 8)$, and also find the co-ordinates of the point of intersection of this line with the yz -plane.

[Agra 77]

The co-ordinates of any point on the line joining the two points are

$$\left(\frac{3\lambda - 2}{\lambda + 1}, \frac{-5\lambda + 4}{\lambda + 1}, \frac{8\lambda + 7}{\lambda + 1} \right)$$

If the point is in the yz -plane, then its x -co-ordinate should be zero.

$$\frac{3\lambda - 2}{\lambda + 1} = 0 \text{ or } 3\lambda - 2 = 0 \quad \therefore \lambda = \frac{2}{3}.$$

Hence the required ratio is $2 : 3$, and putting $\lambda = \frac{2}{3}$ the required point is $(0, \frac{2}{5}, \frac{37}{5})$.

Example 2 From any point $(1, -2, 3)$ lines are drawn to meet the sphere $x^2 + y^2 + z^2 = 4$ and they are divided in the ratio $2 : 3$. Prove that the points of section lie on a sphere, and find its equation.

Let the line through $(1, -2, 3)$ meet the sphere in the point (x', y', z') .

$$\therefore x'^2 + y'^2 + z'^2 = 4. \quad \dots(1)$$

Let (α, β, γ) be the point which divides the join of $(1, -2, 3)$ and (x', y', z') in the ratio $2 : 3$.

$$\alpha = \frac{2x' + 3}{2 + 3}; \quad \therefore x' = \frac{5\alpha - 3}{2},$$

$$\beta = \frac{2y' - 6}{2 + 3}; \quad \therefore y' = \frac{5\beta + 6}{2}$$

$$\gamma = \frac{2z' + 9}{2 + 3}; \quad \therefore z' = \frac{5\gamma - 9}{2}$$

Putting the values of x', y', z' in (1) we get

$$(5\alpha - 3)^2 + (5\beta + 6)^2 + (5\gamma - 9)^2 = 4 \times 4$$

$$\text{or } 25(\alpha^2 + \beta^2 + \gamma^2) - 30\alpha + 60\beta - 90\gamma + 110 = 0$$

$$\text{or } 5\alpha^2 + 5\beta^2 + 5\gamma^2 - 6\alpha + 12\beta - 18\gamma + 22 = 0$$

Hence the points of section (α, β, γ) lie on the curve whose equation is

$$5x^2 + 5y^2 + 5z^2 - 6x + 12y - 18z + 22 = 0,$$

which represents a sphere.

Exercise 2

1. Prove that the three points P, Q, R whose co-ordinates are respectively $(3, 2, -4), (5, 4, -6)$ and $(9, 8, -10)$ are collinear, and find the ratio in which the point Q divides PR . [Agra 80]

Ans 1 : 2

Hint. For collinearity it is easy to see that

$$PQ + QR = PR.$$

Let Q divide PR in the ratio $\lambda : 1$ then the point $Q(5, 4, -6)$ is

$$\left(\frac{9\lambda + 3}{\lambda + 1}, \frac{8\lambda + 2}{\lambda + 1}, \frac{-10\lambda - 4}{\lambda + 1} \right).$$

Comparing the x -co-ordinate of Q

$$\frac{9\lambda + 3}{\lambda + 1} = 5; \quad \therefore 9\lambda + 3 = 5\lambda + 5;$$

$$\therefore 4\lambda = 2 \quad \text{or} \quad \lambda = \frac{1}{2}$$

and this value of λ makes the y and z -co-ordinates of Q to be 4, -6 respectively. Also the required ratio is

$$\lambda : 1 \quad \text{or} \quad \frac{1}{2} : 1 \quad \text{or} \quad 1 : 2.$$

2. Find the ratio in which the line joining the points $(-3, 4, 8)$ and $(5, -6, 4)$ is divided by the xy -plane and find the co-ordinates of the point of intersection of the line with the plane.

[Refer Ex. P. 12]

Ans. 2 : 1 ; $(\frac{2}{3}, \frac{8}{3}, 0)$

3. Find the ratio in which the sphere $x^2 + y^2 + z^2 = 504$ divides the line joining the points $(12, -4, 8)$ and $(27, -9, 18)$.

Ans. $-2 : 3 ; 2 : 3$.

Hint. Find the general co-ordinates of a point on the line joining given points and it must satisfy the equation of the given sphere, thus giving you $9\lambda^2 - 4 = 0$ or $\lambda = \pm \frac{2}{3}$ etc.

4. Show that the three lines joining the middle points of opposite edges of a tetrahedron meet in a point.

$$E \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

The mid. point of opp. edge CD is

$$F \left(\frac{x_3 + x_4}{2}, \frac{y_3 + y_4}{2}, \frac{z_3 + z_4}{2} \right)$$

The mid. point of EF is

$$\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{\Sigma y_1}{4}, \frac{\Sigma z_1}{4} \right)$$

which is the C. G. of tetrahedron. Similarly the join of the mid. points of other pairs of opposite edges have same mid. point and hence they are concurrent.

Example 5. (a) A point P lies on the line whose end points are $A(1, 2, 3)$ and $B(2, 10, 1)$. If z -co-ordinate of P is 7, find its other co-ordinate.

Let P divide in the ratio $\lambda : 1$

$$\therefore P \left(\frac{2\lambda + 1}{\lambda + 1}, \frac{10\lambda + 2}{\lambda + 1}, \frac{\lambda + 3}{\lambda + 1} \right). \text{ But } z\text{-co-ordinate is } 7$$

$$\therefore \frac{\lambda + 3}{\lambda + 1} = 7 \text{ or } \lambda = -2/3$$

Putting for λ we get the point P as $(-1, -14, 7)$

(b) The line joining the points $(1, 8, -1)$ and $(4, -4, 2)$ meet the zx and xy planes at P and Q respectively. Prove that the co-ordinates of P and Q are $(3, 0, 1)$ and $(2, 4, 0)$. [Madras 75]

Example 6 (a). Find the distance of the point $(1, 2, 0)$ from the point where the line joining $A(2, -3, 1)$ and $B(3, -4, -5)$ cuts the plane $2x + y + z = 7$. [Ranchi 75]

Let the line AB cut the plane in P whose co-ordinates can be taken as $\left(\frac{3\lambda + 2}{\lambda + 1}, \frac{-4\lambda - 3}{\lambda + 1}, \frac{-5\lambda + 1}{\lambda + 1} \right)$.

Since it lies on the plane $2x + y + z = 7$, we get $\lambda = \frac{1}{3}$.

Hence the point P is $(7/3, -10/3, -3/3)$ and its distance from $(1, 2, 0)$ is

$$\sqrt{\left(\frac{16}{9} + \frac{256}{9} + 1 \right)} = \frac{1}{3} \sqrt{281}$$

(b) Prove that the plane $ax + by + cz + d = 0$ divides the line joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio

$$-\frac{(ax_1 + by_1 + cz_1 + d_1)}{ax_2 + by_2 + cz_2 + d_2} \quad [\text{Calcutta 74}]$$

Proceed exactly as above taking the ratio as $\lambda : 1$ and the value of λ is found to be as given

(c) Prove that the co-ordinates of the point which divides the line joining $(1, -1, 2)$ and $(2, 3, 7)$ in the ratio $2 : 3$ is $(7/5, 3/5, 4)$. [Madras 77]

Direction-Cosines and Projections

§ 1. Angle between two non-intersecting (i.e. non-coplanar lines) :

We already know what we mean by the angle between intersecting, i.e. coplanar lines. The angle between two non-intersecting lines AB and CD is the angle between two straight lines drawn through any point O parallel to AB and CD .

Hence the angle between AB and CD is the angle POQ between the lines OP and OQ parallel to AB and CD respectively.

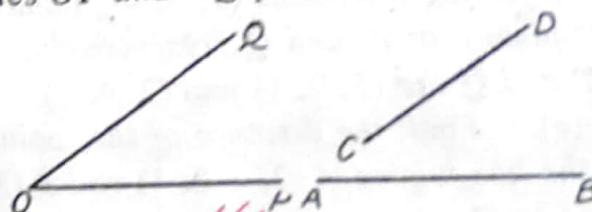


FIG-10

§ 2. Direction cosines of a line.

If a given line AB makes angles α, β, γ , with ~~(the +ve direction of)~~ axes of x, y and z respectively, then $\cos \alpha, \cos \beta, \cos \gamma$ are called the direction-cosines of the line AB and are generally denoted by the letters l, m and n respectively. Again the direction-cosines of BA will be

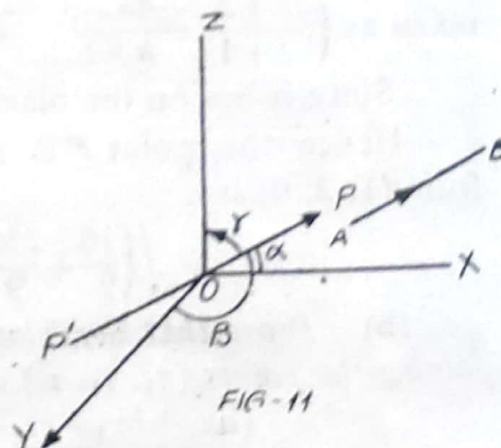


FIG-11

$$\cos (180-\alpha), \cos (180-\beta), \cos (180-\gamma)$$

because the line through O parallel to BA , i.e. OP' will now make angles $180-\alpha, 180-\beta, 180-\gamma$ with the axes.

Note. $\alpha + \beta + \gamma$ is not equal to 360° as these angles do not lie in the same plane.

Direction-cosines (d. c.'s) of the axes of reference.

Now the axis of x makes with the co-ordinate axes the angles

0, 90° , 90° respectively and hence by definition, its direction-cosines are $\cos 0$, $\cos 90^\circ$, $\cos 90^\circ$ i.e. 1, 0, 0.

Similarly the d.c.'s of the axes of y and z are 0, 1, 0, and 0, 0, 1 respectively.

§ 3. To prove that if OP be a line through origin and of length r , then the co-ordinates of the point P are (lr, mr, nr) , where l, m, n are the direction cosines of the line OP .

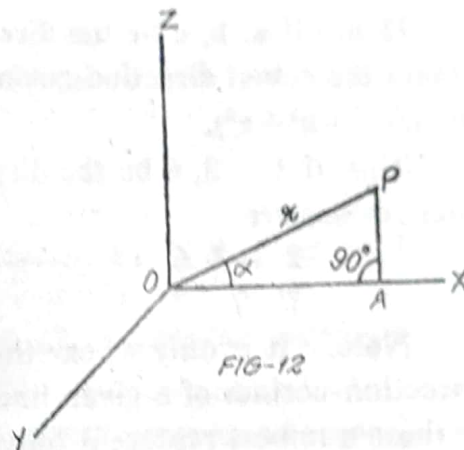
Through P draw PA perpendicular on the axis of x cutting it at the point A . If (x, y, z) be the co-ordinates of P , then $OA = x$, and from right angled $\triangle OAP$,

$$\frac{OA}{OP} = \cos \alpha,$$

or $\frac{x}{r} = l; \therefore x = lr.$

Similarly,

$$y = mr, z = nr.$$



§ 4. To prove that if l, m, n be the direction-cosines of any line AB , then $l^2 + m^2 + n^2 = 1$.

[Burdwan 75, Madras 78, Utkal 78, Bihar 76, Kanpur 81, 76, Gauhati 74]

Through the origin O draw a line OP parallel to AB and of length r so that the direction-cosines of the line OP are l, m, n . Now if (x, y, z) be the co-ordinates of the point P then by § 3, we know that

$$x = lr, y = mr, z = nr.$$

$$\therefore x^2 + y^2 + z^2 = r^2 (l^2 + m^2 + n^2) \quad \dots (1)$$

But

$$x^2 + y^2 + z^2 = OP^2 = r^2$$

Hence from (1), we get $r^2 = r^2 (l^2 + m^2 + n^2)$

or

$$l^2 + m^2 + n^2 = 1$$

or

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1. \quad \dots (2)$$

where α, β, γ are the angles which the line AB and hence OP makes with the axes of x, y and z respectively.

§ 5. Direction-ratios. Any three numbers a, b and c which are proportional to the direction-cosines l, m, n , respectively of given line are called direction-ratios of the given line. Now from the definition of direction-ratios, we have

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}} = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}} \quad (\S 4).$$

$$\therefore l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}},$$

$$n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}.$$

where the same sign either +ive or -ive is to be chosen throughout.

Hence if a, b, c be the direction-ratios of a given line then to obtain the actual direction-cosines we have to divide each of them by $\sqrt{a^2 + b^2 + c^2}$.

Thus if 2, -3, 6 be the direction-ratios, then the actual direction-cosines are

$$\frac{2}{7}, \frac{-3}{7}, \frac{6}{7}$$

$$\because \sqrt{2^2 + (-3)^2 + 6^2} = 7$$

Note. It is only when these numbers represent the actual direction-cosines of a given line that sum of their squares is unity. If these numbers represent only the direction-ratios, then sum of their squares is not equal to unity.

6. Projection of a point.

The projection of a point A on a given line CD is the point B which is the foot of the perpendicular from A on CD . This is the point in which the plane through A and perpendicular to CD meets it.

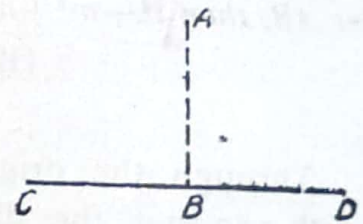


FIG-13

Projection of a segment of a line on another line and its length.

The projection of a given line AB on another line CD is the segment $A'B'$ of CD where A' and B' are the projections of the points A and B on the line CD . In other words the projection of AB on CD is the intercept $A'B'$ made on the line CD by the planes through A and B each perpendicular to CD .

Length of the projection $A'B'$.

Through A draw AP parallel to $A'B'$ and meeting the plane through B in the point P .

$$\therefore AP = A'B'.$$

Now if θ be the angle between the lines AB and CD , then AP being parallel to CD , the angle PAB is also θ . Again PB is any line in the plane which is perpendicular to CD and hence to AP .

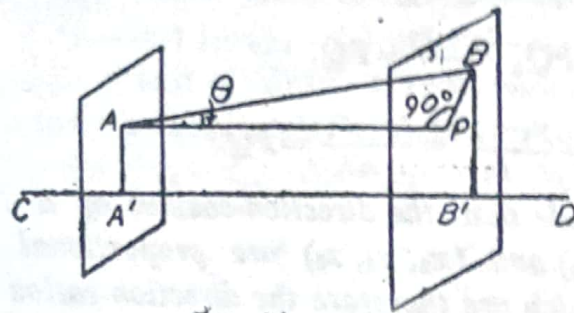


Fig-14

$$\therefore \angle APB = 90^\circ$$

$$\therefore A'B' = AP = AB \cos \theta.$$

Thus the projection of a line AB on another line CD is $AB \cos \theta$ where θ is the angle between the lines AB and CD .

Projection of a broken line consisting of continuous segments on a given line.

Let $A, B, C, D, \dots, M, \dots, N$ be any number of points in space and $A', B', C', D', \dots, M', \dots, N'$ be their projections on the line PQ and these points being on the same straight line, we have

$$A'B' + B'C' + C'D' + \dots + M'N' = A'N'.$$

But $A'B'$ is the projection of AB on PQ .

Hence sum of the projections of $AB, BC, CD, \dots, MN$ on PQ = projection of AN on PQ i.e., $A'N'$.

§7. Direction-cosines of a line joining two points.

$P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

Let P' and Q' be the projection of two points P and Q on the axes of x , and the direction-cosines of the line joining P and Q be l, m, n . If O be the origin, then $OP' = x_1$ and $OQ' = x_2$, so that $P'Q' = OQ' - OP' = x_2 - x_1$. $P'Q'$ is the projection of the line PQ on the axis of x . If PQ makes angles α, β, γ with the axes of x, y and z respectively, then $l = \cos \alpha, m = \cos \beta, n = \cos \gamma$.

Now we know from § 6 that the projection of a line AB on another line CD is $AB \cos \theta$ where θ is the angle between AB and CD .

$$\therefore P'Q' = PQ \cos \alpha = l PQ$$

$$\text{or } \frac{(x_2 - x_1)}{l} = PQ.$$

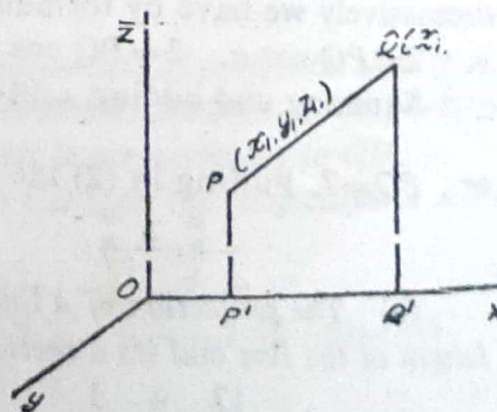


Fig. 15

Similarly by projecting PQ on the axes of y and z , we get

$$\frac{y_2 - y_1}{m} = PQ, \quad \frac{z_2 - z_1}{n} = PQ.$$

$$\therefore x_2 - x_1 = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n} = PQ.$$

Hence we conclude that the direction-cosines of a line joining two points (x_1, y_1, z_1) and (x_2, y_2, z_2) are proportional to $x_2 - x_1$, $y_2 - y_1$ and $z_2 - z_1$ which are therefore the direction ratios of the line PQ and the actual direction-cosines are

$$\frac{x_2 - x_1}{PQ}, \frac{y_2 - y_1}{PQ}, \frac{z_2 - z_1}{PQ},$$

where $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

✓ Ex. : Thus if P be $(3, -4, 4)$ and Q be $(2, -9, 2)$, then the direction ratios of the line PQ are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ i.e. $(2 - 3, -9 - (-4), 2 - 4)$ i.e. $-1, -5, -2$.

$\therefore$ actual direction-cosines of the line PQ are

$$\frac{-1}{\sqrt{1+25+4}}, \frac{-5}{\sqrt{1+25+4}}, \frac{-2}{\sqrt{1+25+4}}$$

$$\text{i.e. } \frac{-1}{\sqrt{30}}, \frac{-5}{\sqrt{30}}, \frac{-2}{\sqrt{30}}.$$

Note. The direction-cosines of the line QP are

$$\frac{1}{\sqrt{30}}, \frac{5}{\sqrt{30}}, \frac{2}{\sqrt{30}}.$$

Example 1 (a) The projection of a line on the co-ordinate axes are 2, 3 and 6. Find the length and the direction-cosines of the line.

[Agra 81, Barhampur 81]

Let the line be PQ and its direction-cosines be l, m, n then we know that $P'Q' = PQ \cos \theta$... (1)

where $P'Q'$ is the projection of PQ on a line AB and θ is the angle between PQ and AB . Choosing AB as x -axis, y -axis and z -axis successively we have by formula § 6 P. 18.

$$2 = PQ \cos \alpha, \quad 3 = PQ \cos \beta, \quad 6 = PQ \cos \gamma \quad \dots (2)$$

$$\text{Squaring and adding } \sqrt{4+9+36} = PQ (\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma)^{1/2}$$

or $PQ = 7$. Putting in (2) the direction-cosines are

$$\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$$

(b) The projection of a line on the axes are 12, 4, 3. Find the length of the line and its direction-cosines [Raj. 73]

$$\text{Ans. } 13; \frac{12}{13}, \frac{4}{13}, \frac{3}{13}.$$

§ 8. Projection of a line joining $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on another line whose direction-cosines are l, m, n is

$$(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n.$$

Through P and Q draw planes parallel to the co-ordinate planes to form a rectangular parallelopiped; then clearly

$$PA = x_2 - x_1, \quad NA = y_2 - y_1, \quad NQ = z_2 - z_1.$$

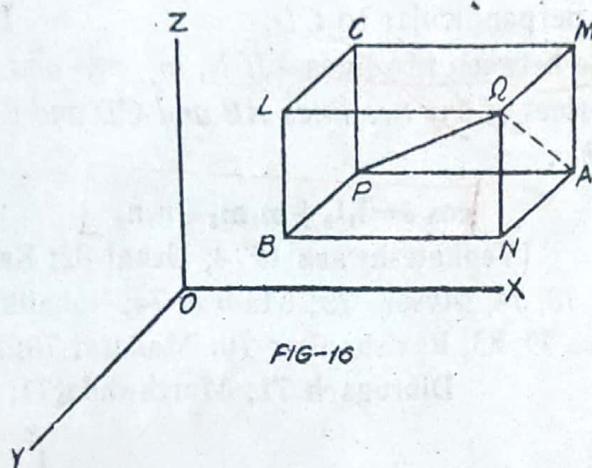


FIG-16

If α, β, γ be the angles which the line whose direction-cosines are l, m, n makes with the axes, then $l = \cos \alpha, m = \cos \beta, n = \cos \gamma$.

Now by § 6 the projection of PA which is parallel to x -axis on that line is $PA \cos \alpha = (x_2 - x_1)l$.

Similarly the projection of AN which is parallel to y -axis on that line is $AN \cos \beta = (y_2 - y_1)m$ and the projection of NQ which is parallel to z -axis on that line is

$$NQ \cos \gamma = (z_2 - z_1)n.$$

Also we know from § 6 that sum of the projections of PA, AN, NQ , on a line is equal to projection of PQ on that line and is therefore equal to

$$(x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n.$$

Hence proved.

Note. You are to take d.c's l, m, n of the line on which projection is made.

Example 2. Prove by the method of projection that if A, B, C, D be the four points whose co-ordinates are $(6, -6, 0), (-1, -7, 6), (3, -4, 4), (2, -9, 2)$, then AB is perpendicular to CD .

If AB is perpendicular to CD then the projection of AB on CD should be zero.

$$A \text{ is } (6, -6, 0), B \text{ is } (-1, -7, 6).$$

$$\therefore x_2 - x_1 = -1 - 6 = -7, y_2 - y_1 = -7 - (-6) = -1,$$

$$z_2 - z_1 = 6 - 0 = 6.$$

Also the d.c.'s of the line CD as calculated in example of § 7 P. 20 are

$$\frac{-1}{\sqrt{(30)}}, \frac{-5}{\sqrt{(30)}}, \frac{-2}{\sqrt{(30)}}$$

Hence by § 8, the projection of AB on CD is

$$=(-7) \left(\frac{-1}{\sqrt{(30)}} \right) + (-1) \left(\frac{-5}{\sqrt{(30)}} \right) + 6 \left(\frac{-2}{\sqrt{(30)}} \right) = 0$$

$\therefore AB$ is perpendicular to CD .

Hence proved

§ 9. Angle between two lines—If l_1, m_1, n_1 , and l_2, m_2, n_2 be the direction-cosines of any two lines AB and CD and θ be the angle between them then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

[Venkateshwara 1974; Utkal 82; Kanpur 82, 77; Ranchi 78, 74; Mysore 73; Madras 74; Rohailkhand 80; 78; Indore 79; 83; Ravishankar 70; Madurai 70; Burdwan 71; Dibrugarh 71; Marthwada 71; Magadh 74]

1st Method

Through the origin O draw two lines OP_1 of length r_1 and OP_2 of length r_2 parallel respectively to the lines with d.c.'s l_1, m_1, n_1 and l_2, m_2, n_2 .

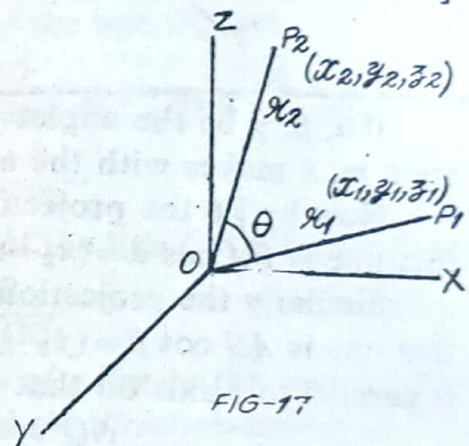


FIG-17

$$\therefore \left. \begin{aligned} x_1 &= l_1 r_1, y_1 = m_1 r_1, z_1 = n_1 r_1 \\ \text{and } x_2 &= l_2 r_2, y_2 = m_2 r_2, z_2 = n_2 r_2 \end{aligned} \right\} (\S 3). \quad \dots (1)$$

If θ be the angle between the given lines, then θ is also equal to the angle between OP_1 and OP_2 .

Now the projection of line OP_2 joining $(0, 0, 0)$ and (x_2, y_2, z_2) on the line OP_1 whose d.c.'s are (l_1, m_1, n_1) is

$$(x_2 - 0) l_1 + (y_2 - 0) m_1 + (z_2 - 0) n_1 = l_1 x_2 + m_1 y_2 + n_1 z_2. \quad \dots (2)$$

But the projection of OP_2 on OP_1 is $OP_2 \cos \theta$. (\S 6)

$$\therefore OP_2 \cos \theta = l_1 x_2 + m_1 y_2 + n_1 z_2$$

$$\text{or } r_2 \cos \theta = l_1 (l_2 r_2) + m_1 (m_2 r_2) + n_1 (n_2 r_2) \text{ from (1).}$$

$$\therefore \cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

2nd Method

We know that in any triangle $a^2 = b^2 + c^2 - 2bc \cos A$.

*

Applying the formula on $\triangle OP_1P_2$, we get

$$P_1P_2^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos \theta \quad \dots(1)$$

$$\text{But } P_1P_2^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

$$= (x_1^2 + y_1^2 + z_1^2) + (x_2^2 + y_2^2 + z_2^2) - 2(x_1x_2 + y_1y_2 + z_1z_2)$$

$$= r_1^2 + r_2^2 - 2r_1r_2 (l_1l_2 + m_1m_2 + n_1n_2) \text{ from (1) of its method.} \quad \dots(2)$$

$$\therefore r_1^2 + r_2^2 - 2r_1r_2 \cos \theta = r_1^2 + r_2^2 - 2r_1r_2 (l_1l_2 + m_1m_2 + n_1n_2)$$

$$\therefore \cos \theta = l_1l_2 + m_1m_2 + n_1n_2 \text{ from (1) and (2).}$$

Corresponding formula when direction-ratios of the two lines be given.

If (a_1, b_1, c_1) and (a_2, b_2, c_2) be the direction-ratios of the two lines, their actual direction-cosines are

$$\left\{ \frac{a_1}{\sqrt{(\Sigma a_1^2)}}, \frac{b_1}{\sqrt{(\Sigma a_1^2)}}, \frac{c_1}{\sqrt{(\Sigma a_1^2)}} \right\}$$

$$\left\{ \frac{a_2}{\sqrt{(\Sigma a_2^2)}}, \frac{b_2}{\sqrt{(\Sigma a_2^2)}}, \frac{c_2}{\sqrt{(\Sigma a_2^2)}} \right\}$$

Hence if θ be the angle between these lines, then

$$\cos \theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}}$$

Cor. 1. Expression for $\sin \theta$ and $\tan \theta$

$$\sin^2 \theta = 1 - \cos^2 \theta = 1 - (l_1l_2 + m_1m_2 + n_1n_2)^2$$

$$= (l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1n_2)^2$$

$$= (l_1^2m_2^2 + l_2^2m_1^2 - 2l_1l_2m_1m_2) + (\dots) + (\dots)$$

$$= (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2$$

$$\therefore \sin^2 \theta = \{(l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2\}$$

$$\text{or } \sin^2 \theta = \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix}^2 + \begin{vmatrix} m_1 & n_1 \\ m_2 & n_2 \end{vmatrix}^2 + \begin{vmatrix} n_1 & l_1 \\ n_2 & l_2 \end{vmatrix}^2$$

In terms of direction-ratios, the above formula takes the form

$$\sin \theta = \frac{\sqrt{\{(a_1b_2 - a_2b_1)^2 + (b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2\}}}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}}$$

Lagrange's Identity.

$$(l_1^2 + m_1^2 + n_1^2)(l_2^2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1n_2)^2$$

$$\equiv (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2$$

The above identity should be committed to memory.

$$\therefore \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sqrt{[\Sigma (l_1m_2 - l_2m_1)^2]}}{\Sigma l_1l_2}$$

in terms of direction-cosines

$$= \frac{\sqrt{[\Sigma (a_1b_2 - a_2b_1)^2]}}{\Sigma a_1a_2}$$

in terms of direction-ratios

We find that the value of $\tan \theta$ is the same whether we use direction-cosines or direction ratios.

Cor. II. Condition for perpendicularity.

[Indore 79 ; Kashmir 75 ; M.D. 79 ; G.N.D. 76 ;
Madras 74 ; Gauhati 67]

(a) In case the lines be perpendicular to each other, then $\theta = 90^\circ$; $\therefore \cos \theta = 0$ and hence the required condition is

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0.$$

If we use direction-ratios, then we get the required condition as

$$\frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{(a_1^2 + b_1^2 + c_1^2)} \sqrt{(a_2^2 + b_2^2 + c_2^2)}} = 0,$$

i.e. $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

Thus we observe that condition of perpendicularity is the same whether we use d.c.'s or direction-ratios.

Condition for parallelism.

[Indore 79, Kashmir 75,
Madras 74, Berhampur 81]

(b) In case the lines be parallel, then

$$\theta = 0 ; \therefore \sin \theta = 0$$

and we get $(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2 = 0$.

The L.H.S. is the sum of three positive expressions and can be zero if each of them is separately zero and hence we get

$$l_1 m_2 - l_2 m_1 = 0, m_1 n_2 - m_2 n_1 = 0, n_1 l_2 - n_2 l_1 = 0$$

or $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{\sqrt{(l_1^2 + m_1^2 + n_1^2)}}{\sqrt{(l_2^2 + m_2^2 + n_2^2)}} = 1$

i.e. $l_1 = l_2, m_1 = m_2, n_1 = n_2$.

i.e. the direction-cosines of the two lines are the same.

If we use the direction-ratios, then from the corresponding formula of $\sin \theta$ for direction-ratios, the condition becomes

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}. \text{ Here } [(a_1^2 + b_1^2 + c_1^2) \neq 1]$$

i.e. the direction-ratios of the two lines are proportional.

Example 3. If A, B, C, D are the points

$A(3, 4, 5), B(4, 6, 3), C(-1, 2, 4), D(1, 0, 5)$, find the projection of CD on AB . [Agra 80 ; Rohailkhand 77]

Also determine the angle between AB and CD .

Here we shall choose $x_2 - x_1, y_2 - y_1, z_2 - z_1$ for CD which are 2, -2, 1,

l, m, n will be chosen for AB , where

DRs of AB are 4-3, 6-4, 3-5 or 1, 2, -2

D.C's of AB are $1/3, 2/3, -2/3$ (1)

$\therefore$ Projection of CD on AB is $\Sigma l(x_2 - x_1)$
 $= 1/3 \cdot 2 + 2/3(-2) - 2/3(1) = -4/3 = 4/3$ numerically

D.C's of CD are $2/3, -2/3, 1/3$

Hence from 1 and 2 the angle between them is given by

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2 = 1/9 (2 \cdot 1 - 2 \cdot 2 + 1 \cdot (-2)) \\ = -4/9 \therefore \theta = \cos^{-1}(-4/9).$$

Exercise 3

~~Q.1~~ Find the direction-cosines of the line which is equally inclined to the axes. [Burdwan 78, Ranchi 77, Gauhati 69, Agra 79]

If the line makes angle α, β, γ with the axes, then

$$\cos \alpha = \cos \beta = \cos \gamma. \therefore \alpha = \beta = \gamma \text{ given}$$

or
$$\frac{l}{1} = \frac{m}{1} = \frac{n}{1} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{1+1+1}}$$

$$\therefore \text{d.c.'s are } \pm \frac{1}{\sqrt{3}}, \quad \pm \frac{1}{\sqrt{3}}, \quad \pm \frac{1}{\sqrt{3}}.$$

~~Q.2~~ If α, β, γ be the angles which a line makes with the positive directions of the axes, then prove that

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2.$$

[Dibrugarh 71; Agra 78, Gauhati 69]

We know $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\therefore 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$$

or
$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2.$$

~~Q.3~~ Show that the join of the points $(1, 2, 3), (4, 5, 7)$ is parallel to the join of the points $(-4, 3, -6), (2, 9, 2)$.

D. Rs. of line AB by $x_2 - x_1, y_2 - y_1, z_2 - z_1$ are 3, 3, 4.

D. Rs. of line CD are 6, 6, 8 which are proportional to 3, 3, 4 and hence the lines are parallel. [R. (b) P. 24]

4. Prove that the line joining the points $(1, 2, 3), (-1, -2, -3)$, is perpendicular to the line joining points $(-2, 1, 5), (3, 3, 2)$.

D. Rs. of line AB are $-2, -4, -6$.

D. Rs. of line CD are 5, 2, -3 .

$$\text{Also } a_1 a_2 + b_1 b_2 + c_1 c_2 = (-2) \cdot 5 + (-4) \cdot 2 + (-6) \cdot (-3) = 0.$$

Hence the lines are perpendicular. [R (a) P. 24]

5. If (l_1, m_1, n_1) and (l_2, m_2, n_2) are the direction-cosines of two lines then, the direction-ratios of another line which is perpendicular to both the given lines are

$$(m_1 n_2 - m_2 n_1), (n_1 l_2 - n_2 l_1), (l_1 m_2 - l_2 m_1).$$

Prove further that if the given lines be perpendicular to each other, then these direction-ratios are the actual direction cosines.

[Bhagalpur 66, Awadh 78; Bundelkhand 79, Jiwaji 66]

Let (l, m, n) be the direction-cosines of the required line ; then since it is perpendicular to both the given* lines, we have

$$ll_1 + mm_1 + nn_1 = 0.$$

$$ll_2 + mm_2 + nn_2 = 0.$$

$$\therefore \frac{l}{m_1n_2 - m_2n_1} = \frac{m}{n_1l_2 - n_2l_1} = \frac{n}{l_1m_2 - l_2m_1}.$$

Thus we get the direction-ratios. The actual direction-cosines are

$$\frac{m_1n_2 - m_2n_1}{\sqrt{[\Sigma(m_1n_2 - m_2n_1)^2]}} \cdot \frac{n_1l_2 - n_2l_1}{\sqrt{[\Sigma(m_1n_2 - m_2n_1)^2]}} \cdot \frac{l_1m_2 - l_2m_1}{\sqrt{[\Sigma(m_1n_2 - m_2n_1)^2]}}.$$

Now $\sqrt{[\Sigma(m_1n_2 - m_2n_1)^2]} = \sin \theta$, where θ is the angle between the lines whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 . In case these lines be perpendicular, i.e., $\theta = 90^\circ$ then $\sin \theta = 1$, i.e.,

$$\sqrt{[\Sigma(m_1n_2 - m_2n_1)^2]} = 1.$$

$\therefore$ the actual direction-cosines are
~~Q. 6.~~ Lines OA, OB, are drawn from O with direction-cosines proportional to $(1, -1, -1)$ and $(2, -1, 1)$. Find the d.c.'s of the normal to the plane AOB.

If l, m, n be the d.c.'s of normal to the plane AOB then it is perpendicular to both OA and OB and hence $l - m - n = 0$ and

$$2l - m + n = 0 \text{ etc.}$$

$$\text{Ans. } \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}} \text{ or } \frac{-2}{\sqrt{14}}, \frac{-3}{\sqrt{14}}, \frac{1}{\sqrt{14}}.$$

~~Q. 7.~~ If $(l_1, m_1, n_1), (l_2, m_2, n_2), (l_3, m_3, n_3)$ be the direction cosines of three mutually perpendicular lines, then find the direction cosines of a line whose direction-ratios are $l_1 + l_2 + l_3, m_1 + m_2 + m_3, n_1 + n_2 + n_3$ and show that the line is equally inclined to the given lines.

(Vikram 84, Avadh 82)

Since the three given lines are mutually perpendicular, we have

$$l_1l_2 + m_1m_2 + n_1n_2 = 0, \quad l_2l_3 + m_2m_3 + n_2n_3 = 0$$

$$l_3l_1 + m_3m_1 + n_3n_1 = 0.$$

$$\text{Also } \left. \begin{aligned} l_1^2 + m_1^2 + n_1^2 &= l_2^2 + m_2^2 + n_2^2 = l_3^2 + m_3^2 + n_3^2 = 1 \end{aligned} \right\} \dots (1)$$

$$\text{Now } \sqrt{[(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2]} \\ = \sqrt{[\Sigma l_1^2 + \Sigma l_2^2 + \Sigma l_3^2 + 2\Sigma l_1l_2 + 2\Sigma l_2l_3 + 2\Sigma l_3l_1]} = \sqrt{3} \text{ from (1)}$$

Hence the direction cosines are

$$\left(\frac{l_1 + l_2 + l_3}{\sqrt{3}}, \frac{m_1 + m_2 + m_3}{\sqrt{3}}, \frac{n_1 + n_2 + n_3}{\sqrt{3}} \right) \dots (2)$$

If θ be the angle between the lines whose direction-cosines are (l_1, m_1, n_1) and those given by (2), then

$$\begin{aligned}\cos \theta &= \frac{l_1(l_1+l_2+l_3)}{\sqrt{3}} + \frac{m_1(m_1+m_2+m_3)}{\sqrt{3}} + \frac{n_1(n_1+n_2+n_3)}{\sqrt{3}} \\ &= \frac{\Sigma l_1^2 + \Sigma l_1 l_2 + \Sigma l_1 l_3}{\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ by (1); } \therefore \theta = \cos^{-1} \frac{1}{\sqrt{3}}.\end{aligned}$$

Similarly we can show that the angle between each of the other two lines and the line with d.c.'s given by (2) is also $\cos^{-1} \frac{1}{\sqrt{3}}$.

8. Prove that three concurrent lines with direction cosines (l_1, m_1, n_1) , (l_2, m_2, n_2) and (l_3, m_3, n_3) are coplanar if

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0.$$

(Rohailkhand 1974 ; Madras 75 ; Meerut 75 ;
Rawa 74 ; Agra 77, 74 ; Indore 70)

Let (l, m, n) be the direction cosines of a line which is normal to the plane that contains the two lines whose d.c.'s are l_1, m_1, n_1 and l_2, m_2, n_2 ; then

$$ll_1 + mm_1 + nn_1 = 0, \quad \dots(1)$$

$$ll_2 + mm_2 + nn_2 = 0. \quad \dots(2)$$

Normal to a plane is perpendicular to every line lying in that plane.

Again if the line whose d.c.'s are (l_3, m_3, n_3) also lies in that plane, then the line with d.c.'s (l, m, n) i.e. normal is also perpendicular to it.

$$\therefore ll_3 + mm_3 + nn_3 = 0 \quad \dots(3)$$

Eliminating l, m, n between (1), (2) and (3) we get the required condition in determinant form as given.

(b) The d.c.'s of two intersecting lines are l_1, m_1, n_1 and l_2, m_2, n_2 . Show that all lines through their intersection and whose d.c.'s are proportional to $l_1 + \lambda l_2, m_1 + \lambda m_2, n_1 + \lambda n_2$ are coplanar with them.

Let l, m, n be the d.c.'s of the normal to the plane containing the given lines so that it is perpendicular to both of them. Hence

$$ll_1 + mm_1 + nn_1 = 0 \quad \dots(1)$$

$$ll_2 + mm_2 + nn_2 = 0 \quad \dots(2)$$

Multiplying (2) by λ and adding to (1) we have

$$l(l_1 + \lambda l_2) + m(m_1 + \lambda m_2) + n(n_1 + \lambda n_2) = 0$$

Above relation shows that the line l, m, n is also perpendicular to the lines whose D.R.'s are $l_1 + \lambda l_2, m_1 + \lambda m_2, n_1 + \lambda n_2$. Therefore the line also lies in the same plane in which the given two lines lie.

9. Prove that three lines drawn from O whose direction-cosines are proportional to $(1, -1, 1), (2, -3, 0)$ and $(1, 0, 3)$ are coplanar.

✓ 10. If the edges of a rectangular parallelepiped be a, b, c show that the angles between the four diagonals are given by

$$\cos^{-1} \frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}.$$

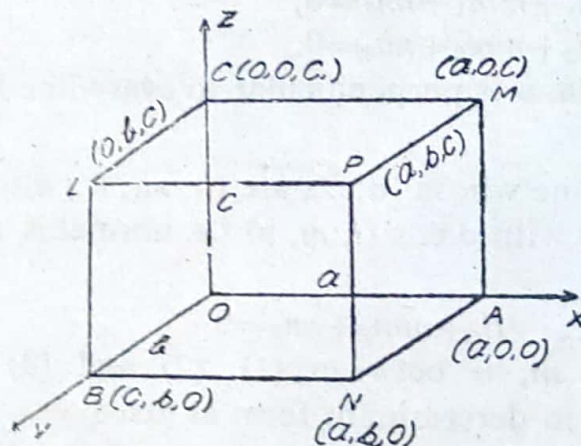
(Kanpur 1982, 77; Meerut 78, 84 (P); Sambalpur 74; Rohailkhand 82; Garhwal 78]

The four diagonals are AL, BM, CN and OP where the co-ordinates of O are $(0, 0, 0)$ $P(a, b, c)$ $A(a, 0, 0)$ $B(0, b, 0)$, $C(0, 0, c)$, $L(0, b, c)$, $M(a, 0, c)$ and $N(a, b, 0)$ [See Ex. P. 4].

Direction ratios of OP are $a-0, b-0, c-0$, i.e. a, b, c .

$$\therefore \text{d.c.'s are } \frac{a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}}$$

and D.Rs.' of AL are $0-a, b-0, c-0$, i.e. $-a, b, c$.



$$\therefore \text{d.c.'s are } \frac{-a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}}$$

Similarly the direction-cosines of BM and CN are

$$\frac{a}{\sqrt{(\Sigma a^2)}}, \frac{-b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}} \text{ and } \frac{a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{-c}{\sqrt{(\Sigma a^2)}} \text{ respectively.}$$

If θ be the angle between OP and AL , we get

$$\cos \theta = \frac{a(-a) + b(b) + c(c)}{\sqrt{(a^2 + b^2 + c^2)}\sqrt{(a^2 + b^2 + c^2)}} = \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2} \dots (1)$$

Similarly we can find the angle between other pairs and hence the angles in general between the six pairs are given by

$$\cos^{-1} \frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}.$$

Note. The ambiguous signs cannot be either all +ive or -ive for in that case $\theta = \cos^{-1} 1$ or $\cos^{-1} -1$, i.e., $\theta = 0$ or 180° which is impossible as the diagonals are not parallel.

11. Prove that the angle between two diagonals of a cube is $\cos^{-1} \frac{1}{3}$.
(Burdwan 1977 ; Berhampore 81 ; Indore 82 ; Calicut 82 ; Vikram 78 ; Gauhati 74)

Cube is a rectangular parallelopiped whose edges a, b, c are all equal. If θ be the angle between diagonals OP and AL then putting $a=b=c$ in (1) of Q. 10.

$$\cos \theta = \frac{-1 + 1 + 1}{1 + 1 + 1} = \frac{1}{3}; \therefore \theta = \cos^{-1} \left(\frac{1}{3} \right).$$

12. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube; show that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3},$$

[Indore 1983, 84 ; Venka 84 ; L.N.M. 82 ; Ranchi 83 ;

Madras 78, 76 ; Lucknow 80 ; Bhagalpur 75 ;

Sambalpur 73 ; Meerut 83, 80, 77 ; Kanpur 81 ;

Garhwal 81, 83 ; Gorakhpur 82]

From Q. 10, d.c.'s of the four diagonals of a cube on putting $a=b=c$ are

$$OP \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), AL \left(\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

$$BM \left(\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), CN \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right)$$

Let l, m, n be d.c.'s of a line which is inclined at angles $\alpha, \beta, \gamma, \delta$, respectively to the four diagonals ; then

$$\cos \alpha = l \cdot \frac{1}{\sqrt{3}} + m \cdot \frac{1}{\sqrt{3}} + n \cdot \frac{1}{\sqrt{3}} = \frac{l+m+n}{\sqrt{3}}.$$

$$\text{Similarly } \cos \beta = \frac{-l+m+n}{\sqrt{3}}, \cos \gamma = \frac{l-m+n}{\sqrt{3}},$$

$$\cos \delta = \frac{l+m-n}{\sqrt{3}}.$$

$$\begin{aligned} \therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta \\ = \frac{1}{4} [(l+m+n)^2 + (-l+m+n)^2 + (l-m+n)^2 + (l+m-n)^2] \\ = \frac{1}{4} \cdot 4 (l^2 + m^2 + n^2) = \frac{4}{4} \end{aligned}$$

13. (a) If (l_1, m_1, n_1) and (l_2, m_2, n_2) are the d.c.'s of two concurrent lines, show that d.c.'s of two lines bisecting the angle between them are proportional to

$$l_1 \pm l_2, m_1 \pm m_2, n_1 \pm n_2.$$

In case the angle between the given lines be θ then find the direction-cosines of these bisectors [Rajasthan 78; Garhwal 78]

Through the origin O draw two lines parallel to given lines and mark off points P and Q on them at distances r each so that

P is $(l_1 r, m_1 r, n_1 r)$

and Q is $(l_2 r, m_2 r, n_2 r)$

If A be the mid point of PQ then A is

$$\left\{ \frac{1}{2} (l_1 + l_2) r, -, - \right\}$$

Hence OA will be the internal bisector where O is $(0, 0, 0)$ clearly DR's of OA are $\left\{ \frac{1}{2} (l_1 + l_2) r, 0, 0 \right\}$ or

$$l_1 + l_2, m_1 + m_2, n_1 + n_2. \quad \dots(1)$$

Again if we produce QO and choose a point R at a distance r then co-ordinates of R are $(-l_2 r, -m_2 r, -n_2 r)$

Now if B be the mid point of PR then

$$B \text{ is } \left\{ \frac{1}{2} (l_1 - l_2) r, -, - \right\}$$

Hence OB will be the external bisector where O is $(0, 0, 0)$ DR's of OB are

$$l_1 - l_2, m_1 - m_2, n_1 - n_2 \quad \dots(2)$$

Again if θ be angle between the lines then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2 \quad \dots(3)$$

$$\begin{aligned} \text{Now } \Sigma(l_1 + l_2)^2 &= \Sigma l_1^2 + \Sigma l_2^2 + 2 \Sigma l_1 l_2 = 1 + 1 + 2 \cos \theta \\ &= 2(1 + \cos \theta) = 4 \cos^2 \theta/2 \end{aligned}$$

$$\text{and } \Sigma(l_1 - l_2)^2 = 2(1 - \cos \theta) = 4 \sin^2 \theta/2$$

Hence from 1 and 2 the DC's of bisectors are

$$\frac{l_1 + l_2}{2 \cos \theta/2}, \frac{m_1 + m_2}{2 \cos \theta/2}, \frac{n_1 + n_2}{2 \cos \theta/2} \quad \text{and}$$

$$\frac{l_1 - l_2}{2 \sin \theta/2}, \frac{m_1 - m_2}{2 \sin \theta/2}, \frac{n_1 - n_2}{2 \sin \theta/2}.$$

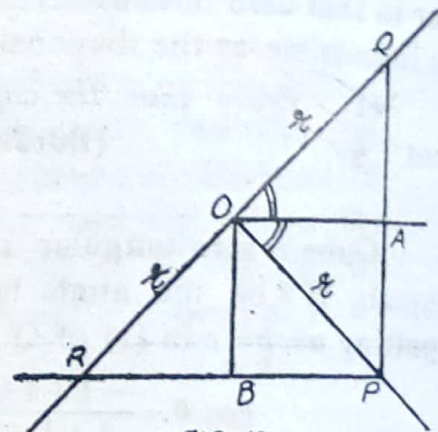


FIG-19

(b) If $l_1, m_1, n_1; l_2, m_2, n_2$ are two directions inclined at an angle θ to each other, show that the directions

$$\frac{l_1+l_2}{2 \cos \theta/2}, \frac{m_1+m_2}{2 \cos \theta/2}, \frac{n_1+n_2}{2 \cos \theta/2}$$

bisect the angle between these lines.

14. The direction-cosines of a moving line in two adjacent positions are l, m, n and $l+\delta l, m+\delta m, n+\delta n$. Show that the small angle $\delta\theta$ between the positions is given by

$$(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2. \quad [\text{Garhwal 75; Raj. 83}]$$

$$\Sigma l^2 = l^2 + m^2 + n^2 = 1 \quad \dots (1)$$

$$\Sigma (l+\delta l)^2 = (l+\delta l)^2 + (m+\delta m)^2 + (n+\delta n)^2 = 1 \quad \dots (2)$$

Also

$$\cos \delta\theta = \Sigma l(l+\delta l) = l(l+\delta l) + m(m+\delta m) + n(n+\delta n) \quad \dots (3)$$

$$\text{Now} \quad \delta l = l + \delta l - l;$$

$$\therefore (\delta l)^2 = (l + \delta l)^2 + l^2 - 2l(l + \delta l)$$

$$\therefore \delta l^2 + \delta m^2 + \delta n^2 = \Sigma (l + \delta l)^2 + \Sigma l^2 - 2 \Sigma l(l + \delta l)$$

$$= 1 + 1 - 2 \cos \delta\theta \text{ by (1), (2) and (3)}$$

$$= 2(1 - \cos \delta\theta) = 2 \cdot 2 \sin^2 \frac{\delta\theta}{2}$$

$$= 4 \left(\frac{\delta\theta}{2} \right)^2 = \delta\theta^2;$$

$\therefore$ where α is small then $\sin \alpha = \alpha$.

15. Prove that the area of triangle OAB where O is origin and A and B are the points (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively is

$$\frac{1}{2} \sqrt{[(y_1 z_2 - y_2 z_1)^2 + (z_1 x_2 - z_2 x_1)^2 + (x_1 y_2 - x_2 y_1)^2]}$$

[Rohailkhand 81; Jodhpur 76; Kerala 69]

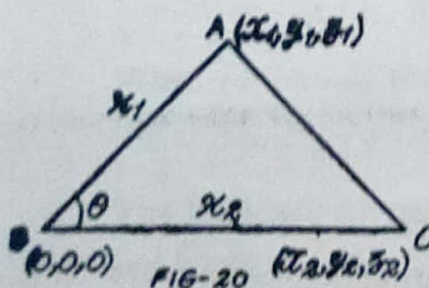
Area of triangle OAB is

$$\frac{1}{2} r_1 r_2 \sin \theta \quad \dots (2)$$

$$\text{where } r_1 = \sqrt{(x_1^2 + y_1^2 + z_1^2)}$$

$$\text{and } r_2 = \sqrt{(x_2^2 + y_2^2 + z_2^2)}$$

Again direction-ratios of OA and OB are (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively



$$\sin \theta = \frac{\sqrt{[\Sigma (y_1 z_2 - y_2 z_1)^2]}}{\sqrt{[\Sigma (x_1^2) \Sigma (x_2^2)]}} \quad [\text{P. 29}]$$

Hence putting the values of r_1, r_2 and $\sin \theta$ in (1), we get the required result.

100% 16. (a) Prove that the acute angle between the lines whose direction-cosines are given by the relations

$l+m+n=0$ and $l^2+m^2+n^2=0$ is $\pi/3$.

[Punjab Supplementary 1943]

Eliminating n , we get

$$l^2+m^2-(l+m)^2=0; \therefore 2lm=0,$$

When $l=0$ then $m+n=0$ or $m=-n$;

$$\therefore \frac{l}{0} = \frac{m}{1} = \frac{n}{-1} \quad \dots(1)$$

When $m=0$ then $l+n=0$ or $l=-n$;

$$\therefore \frac{l}{1} = \frac{m}{0} = \frac{n}{-1} \quad \dots(2)$$

Hence the D.R.'s of the lines are $0, 1, -1$; $1, 0, -1$,

100% $\therefore \cos \theta = \frac{0 \cdot 1 + 1 \cdot 1 + 0 \cdot (-1)(-1)}{\sqrt{(0+1+1)} \cdot \sqrt{(0+1+1)}} = \frac{1}{2}; \therefore \theta = \pi/3,$

✓ (b) The direction-cosines of two lines are connected by the relations $l-5m+3n=0$ and $7l^2+5m^2-3n^2=0$ Find them

[Raj. 81; Madras 74]

Eliminating n between the given relations we get

$$7(5m-3n)^2+5m^2-3n^2=0$$

$$180m^2-210mn+60n^2=0$$

$$6m^2-7mn+2n^2=0$$

$$(3m-2n)(2m-n)=0$$

$$\therefore \frac{m}{n} = \frac{2}{3} \text{ and hence } \frac{l}{n} - \frac{5m}{n} + 3 = 0$$

$$\text{or } \frac{l}{n} = \frac{5 \cdot 2}{3} - 3 = \frac{1}{3} \quad \frac{l}{1} = \frac{m}{2} = \frac{n}{3}$$

$$\text{If } \frac{m}{n} = \frac{1}{2} \text{ then } \frac{l}{n} = 5 \cdot \frac{1}{2} - 3 = -\frac{1}{2}$$

$$\frac{l}{-1} = \frac{m}{1} = \frac{n}{2}$$

The direction-ratios of the two lines are

$1, 2, 3$ and $-1, 1, 2$. Hence direction-cosines are

$$\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \text{ and } \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$$

100% (c) Prove that the angle between the lines whose direction cosines are connected by the relations $l+m+n=0$ and

$$2lm+2nl-mn=0 \text{ is } 2\pi/3$$

[Meerut 83; Kurkshetra 73; Agra 74, Bihar 74]